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Integrating digital influencer persuasion model and theory of planned behavior: The mediating role of consumer involvement in endorsed brands

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ABSTRACT

This study explores the integration of the Digital Influencer Persuasion (DIP) model with the Theory of Planned Behavior (TPB) to examine customer buying behavior. It also investigates the mediating role of consumer involvement in linking persuasion factors to attitudes towards the endorsed brand. Data was collected from 322 valid participants using a survey study approach, and the partial least squares-structural equation modeling (PLS-SEM) technique was employed to test the structural model rigorously. The findings indicate that perceived influence and trustworthiness significantly enhance customer involvement with endorsed brands, whereas reputation does not exhibit a significant effect. Furthermore, customer involvement serves as a strong predictor of customer attitudes toward endorsed brands. While customer attitudes and perceived behavioral control significantly influence purchase intention, subjective norms do not demonstrate a significant effect. Additionally, purchase intention is found to be a significant determinant of actual purchase behavior. Importantly, customer involvement in endorsed brands positively mediates the relationship between attitude toward endorsed brands and key persuasion model factors (i.e., perceived influence and trustworthiness). This study offers valuable theoretical and practical insights by expanding the understanding of how DIs' perceived influence, trustworthiness, and reputation collectively shape consumer behavior. The findings also provide marketers with guidance on selecting suitable DIs who align with their target audience and brand objectives.

1. Introduction

As a critical influencing factor, reference groups can play a crucial role in changing the attitudes and behavioral intentions of both actual and potential buyers. Reference groups can comprise family, friends, or any individual with knowledge and expertise about a product, brand, or service (Pérez-Torres, 2024; Schiffman et al., 2013). The recent advancements in consumer technology have transformed reference groups into digital influencers (DIs), and the diffusion of multiple digital platforms has further strengthened this emerging concept. DIs can be characterized as individuals having huge fan bases (including mega, macro, micro, and nano influencers) developed through valuable and innovative content generation on numerous digital platforms, and can act as influencers for their followers (Geyser, 2022). DIs, also known as social media influencers (SMIs), strategically allow brands to maximize their digital reach and impact by amplifying electronic word-of-mouth (eWOM) and substantially improving brand communication on digital media (Grgurić Čop et al., 2024; Zhou et al., 2021). Thus, DIs are becoming popular for marketers to reach an extensive number of actual and potential customers and can bring real sales, specifically from millennials and the Z generation. Hence, marketers are switching their advertising budget to DIs. According to Statista (2023) the global market size for DIs reached 21.1 billion dollars, nearly tripling its value in 2019. Further, advertising spending on influencer marketing will reach $7.14

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billion by 2024, from $3.9 billion in 2021. Hence, based on the above arguments, it can be presumed that marketers are investing and promoting their brands through influencer marketing to communicate with their customers. Therefore, there is a need to have a holistic understanding of the dynamics of persuasive communication in the contexts of DIs and consumer behavior (Masuda et al., 2022).

DIs or SMIs have been investigated by previous studies across various contexts, including the impact of social media micro and macro influencers on consumer word-of-mouth (WOM) intention (Li et al., 2024) and customer engagement level elicited through product endorsements on Instagram by social-media influencers (Silva et al., 2020). Similarly, Myers et al. (2024) contemplated the influence of pet influencers' non-entertaining promotional posts (visual and textual cues) on customer behavioral engagement. Employing the Uses and Gratification (UGT) theory and Persuasion Knowledge model, the authors found that the endorsement of all these textual and visual cues by pet influencers resulted in diminished customer behavioral engagement. Likewise, Koles et al. (2024) explored the concept of authenticity in the realm of virtual influencers, focusing primarily on entity-referent correspondence authenticity. The study results revealed three different authenticity manifestations (true to ideal, true to self, and true to fact) in the domain of virtual influencers. Moreover, other scholars have examined the impact of digital virtual influencers on consumer decision-making for a certain product in comparison to human digital influencers (Belanche et al., 2024) and individual responses and reactions to sustainable and pro-environmental behavior promoted by virtual influencers (Gerrath et al., 2024).

Alcántara-Pilar et al. (2024) accentuated the role of SMIs in the sphere of digital marketing and examined the effect of influencer trust and credibility on marketing campaigns, consumer loyalty, recommendation or referral behavior, and purchase intention in the TikTok context by integrating stimulus-organism-response and commitment-trust frameworks. The extensive literature review reveals that the existing literature considers the prevalent role of DIs in different domains and identifies their crucial role in brand promotion, customer communication through different digital platforms, customers' emotional expressions, and increased sales and purchase intention (Masuda et al., 2022; Rhee et al., 2021; Shuqair et al., 2024).

DIs have emerged on social media as powerful opinion leaders who engage their followers through compelling content and brand endorsements (Casaló et al., 2020). Research indicates that a DI's perceived influence significantly affects brand engagement, purchase intention, and the likelihood of recommending products. Although some studies have focused solely on perceived influence (Jiménez-Castillo and Sánchez-Fernández, 2019), prior literature suggests that DIs should be evaluated across multiple attributes—such as trustworthiness, reputation, and involvement—to fully capture their impact on consumer perceptions and behaviors (Bastrygina et al., 2024; Joshi et al., 2023; Lou and Yuan, 2019).

Many DI marketing studies have applied persuasion knowledge models that concentrate on consumers' understanding of marketer's motives and tactics, thereby aiding them in recognizing the rationale behind brand endorsements (Friestad and Wright, 1994; Myers et al., 2024). However, it remains unclear how elements from persuasion theory—like trustworthiness, reputation, and perceived influence—integrate with the TPB (Ajzen, 1991) to explain consumer buying behavior. In addition, the mediating role of customer involvement (CI) with endorsed brands in linking influencers' attributes to consumer attitudes warrants further investigation.

Given the significant role DIs play in shaping consumer decisions, and considering that influencer marketing is subject to budgetary constraints (Powderly, 2022), this study addresses the existing knowledge gap by proposing a Digital Influencer Persuasion (DIP) model. Drawing on the persuasion theory of Hovland et al. (1953) and extending the TPB framework, the DIP model aims to examine how DIs' attributes (i.e., perceived influence, trustworthiness, and reputation) affect CI, which in turn shapes attitudes, intentions, and actual purchase behavior. Moreover, the model investigates the mediating role of CI in linking influencer attributes to consumer attitude towards brand endorsements.

This study enhances the DIP model by examining how perceived influence, trustworthiness, and reputation impact CI in endorsed brands. Further, CI is argued to shape attitudes toward an endorsed brand. Finally, perceived behavioral control (PBC), attitude, and subjective norms, as part of the TPB, are argued to shape perceived intentions, which subsequently may shape the customer's actual behavior toward the brand. The DIP model of this study is important for several reasons. Firstly, practitioners investing in DIs are usually questioned about gaining a return on such an investment (Powderly, 2022). This study presents the DIP model to assist practitioners on whether such an investment supports shaping consumer intentions and behavior. Secondly, the effect of an influencer may be considered as a significant contributor to CI once the influencer has trustworthiness and a good reputation. Accordingly, this study, focusing on digital platforms, evaluates the role of trustworthiness, perceived influence, and reputation in customer involvement (de Brito Silva et al., 2025), addressing gaps left by previous scholars.

Thirdly, on a digital platform, the relationship exists at the customer-to-customer level. Accordingly, these might be subjective norms where peers might influence intentions. In addition, consumer-perceived behavioral control might also be the reason for influencing purchase intentions. Therefore, the DIP model in this study also accounts for the influence of subjective norms and perceived behavioral control (PBC) on purchase intentions, as well as the effect of purchase intention on actual buying behavior, expanding upon the TPB (Ajzen, 1991). Fourthly, with insights from Hovland's persuasion model and extending the TPB, the DIP model aims to provide a comprehensive framework for understanding how DIs use source, message, and audience factors to shape customer attitudes and behaviors effectively. Finally, youth have more access to digital platforms. Accordingly, brand practitioners can figure out ways to approach these young consumers on digital platforms. Asian countries have a greater proportion of the young population (Adamu et al., 2023). These young populations can be the customers with greater lifetime value for the firms. Hence, this study considers an Asian country, Pakistan, where the density of youth is higher, to test the DIP model of this study.

2. Literature review

This study presents a complex DIP model by considering perceived influence, trustworthiness, and reputation impact on customer involvement to shape attitudes toward an endorsed brand. Finally, perceived behavioral control (PBC), attitude, and subjective norms shape perceived intentions and the customer's actual behavior toward the brand. Accordingly, a multiple-theory approach is considered to frame this model. In the next section, each theory and the relation of these theories with the DIP model is elaborated.

2.1. The persuasion theory

The linkage between trustworthiness, reputation, perceived influence, and customer influence in the DIP model can be justified by considering the Persuasion Theory. According to Hovland et al. (1953), persuasion theory is rooted in psychology, communication studies, and sociology, and elucidates the principles that underlie the process of influencing others' attitudes, beliefs, and behaviors (Sparks et al., 2013). Persuasion theory can be functional and beneficial to use in several marketing scenarios, claiming persuasion as an information-sharing process in a specific context to influence the attitudes and behaviors of the intended recipient (Hovland et al., 1953). Persuasion theory offers reasonable theoretical grounds for the present study, highlighting digital influencers as a key source of persuasive communication (Fang et al., 2023). The Digital Influencer Persuasion (DIP) Model, like Hovland

et al.'s (1953) persuasion model, emphasizes the fundamental components that drive changes in customer attitudes and behavioral intentions. Hovland's persuasion model includes the persuader, persuasion situation, persuasive information, and object.

In the proposed DIP model, perceived influence is associated with the source characteristics (persuasive information) emphasized by the Hovland persuasion model. Digital influencers can exert influence based on their credibility, expertise, and attractiveness, and are assumed to be the major drivers affecting marketing efforts (Fang et al., 2023). Trustworthiness relates to the quality of persuasive information, as trustworthy messages are more persuasive. Once an influencer establishes credibility and trust with their audience, it acts as a catalyst, significantly amplifying the impact of their content and messages and helping consumers to better perceive the product value (Cheung et al., 2008). Reputation aligns with the contextual factors (persuasion situation) that can enhance or hinder message reception. In Hovland's model, a favorable persuasion situation can amplify the impact of the message, such as how an influencer's positive reputation boosts the effectiveness of the content. Consumer involvement corresponds to the persuasion object in Hovland's model, which is very crucial, and once consumers are more involved with the object, they are more likely to be affected by an influencer's perceived influence, trustworthiness, and reputation. These factors, in turn, significantly impact customers' attitudes toward an endorsed brand.

2.2. The theory of Planned Behavior

In the DIP model, the linkage between attitudes, subjective norms, PBC, and perceived intentions, while the linkage between intentions and actual customer behavior can be argued with the TPB (Ajzen, 1991). The TPB was proposed by Ajzen (1985), which states that individual beliefs and behavior, including attitude, subjective norms, and PBC, lead to behavioral intention and, subsequently, to actual behavior (Ajzen, 1991). The TPB explains that behavior is followed by individual intentions. Thus, attitude, subjective norms, and PBC resolve the behavioral intention leading to actual individual behavior (Ajzen, 1985). Briefly, PBC comprises the individual's conviction to control their behavior, such as having sufficient abilities, skills, and resources to realize the behavior (Ajzen, 1985). Subjective norm refers to the perceived (dis-)approval of social peers, for example, family, friends, and social networks, of a particular decision (Ajzen, 1991). Social influence has a substantial function in modeling consumer buying behavior (Bhukya and Paul, 2023). Lastly, attitude represents the individual's evaluation of how (un-) favorable the intended behavior might be (Ajzen, 1985).

In other words, attitude is a personal evaluation of performing the intended behavior. Previous studies have considered TPB in different domains, such as digital marketing, advertising (Buchanan et al., 2018; Harb et al., 2019), social media (Zahrai et al., 2022), and tourism (Zahrai et al., 2022). In this current study, the TPB is extended by considering additional digital media constructs in the context of DIs. Hence, this investigation proposes and empirically examines the impact of trustworthiness, reputation, and perceived influence on CI in endorsed brands. In contrast, CI in endorsed brands is argued to be an antecedent of DIs' attitude towards the endorsed brand on social media.

2.3. Customer involvement (CI)

In marketing, CI can be considered an important concept. This concept is gaining an excessive amount of interest from scholars and practitioners due to its applicability in the field of behavioral studies. For example, social psychology, advertising, marketing, and tourism (Abbasi et al., 2023; Dedeoğlu, 2019). CI is a pivotal and evolving research niche in the field of consumer behavior and marketing (Altinay and Taheri, 2019). Initially, Zaichkowsky (1985) regarded CI as a person's relevance to an object due to their essential intrinsic needs, values, and interests. Later on, various scholars conceptualized CI as the motivational condition influencing consumers' attention and ability to comprehend (Ghali-Zinoubi and Toukabri, 2019). The significance of CI in the persuasive message also applies to digital media (Mao and Zhang, 2017). This study draws on the work of (McClure and Seock, 2020; Zaichkowsky, 1985) and considers the term "involvement" as a consumer's willingness to engage with a brand endorsed by DIs on social media platforms. The importance of CI in association with attitudes, purchase intentions, and information-searching behavior has been confirmed in prior studies (Ko and Park, 2002; Tata et al., 2019). Thus, the current study tests the impact of particular DIP attributes on CI.

2.4. Digital influencer (DI)

The marketing and advertising descriptions in the literature have established that celebrity endorsement promotes attitudes toward brands and increases purchase intention (Torres et al., 2019). The proliferation of online platforms, cyberspace's scalability, and the speedy diffusion of digitalization have enabled a new kind of influencer, currently known as the DI. A DI or social media influencer is a content creator (Vlogger or Bloggers) who has expertise in a certain area and has accumulated a mass fan base of audience by posting valuable and creative content including Vlogs or Blogs on online media platforms (De Veirman et al., 2017) and act as an influencer in their audience (Delbaere et al., 2021). DIs engage with their followers by continuously uploading content on media (Shuqair et al., 2024). The DIs have the power to switch the buying decisions of the followers due to their position, authority, knowledge, or close attachment with their followers (Geyser, 2022). DIs' activities include vlogging, blogging, or creating quick content that gives their viewers and followers insights into their lives, experiences, and opinions (Torres et al., 2019). Hence, the efficacy of DI marketing was reinforced in prior studies, which acknowledged that DI marketing endorses a major extent of credibility that traditional endorsement ordinarily finds hard to attain (Jarrar et al., 2020).

The connectivity between DI and social media users (both followers and non-followers) is likely completed by various digital platforms, such as content communities (e.g., YouTube), SNSs (e.g., Facebook), and microblogging sites (e.g., Twitter), which assist social networks in practicing shared interests and values (Rhee et al., 2021). Some commonly defined classifications of influencers based on the sum of followers and their views on communication platforms are YouTubers, Bloggers, Facebook Vloggers, Instagram Vloggers/Bloggers, Twitterers, Snapchaters, or TikTokers (Wielki, 2020).

Marketers place a high value on DIs because they can promote customer-brand interactions across product categories (Lou and Yuan, 2019). Consumers trust DIs and follow their content on digital media, which helps them obtain information and recommendations related to a particular product/brand (Delbaere et al., 2021). The DIs' recommendation of a specific product effectively triggers customers' choices and behaviors (Silva et al., 2020). Therefore, the existing research suggests that DIs may help brands connect with consumers on three levels: cognition, attachment, and activation (Delbaere et al., 2021). The high involvement action deepens user engagement in the shared content (Freberg et al., 2011) and develops positive attitudes of followers to buy the endorsed products. This makes them an extremely valuable resource.

The growing usage of DIs as an essential marketing practice has increased research interest in investigating the mechanism of inspiring consumers to adopt influencers as a social default (Ki et al., 2022). Researchers have suggested a varied range of influencer variables and categorized them as DIs characteristics (likeability, perceived credibility, trustworthiness, expertise, attractiveness, familiarity), psychologically related influential factors (homophily, parasocial relationship, wishful identification), and content attributes (Vrontis et al., 2021). Ki et al. (2022) examined the impact of content-determined features (perceived attractiveness and credibility) and personality-determined features (perceived closeness and interactivity), and found that these

constructs inspire consumers to perceive influencers as their social defaults, and this impacts their behavioral and emotional response. Besides, some other studies also explained the significance of social media influencer characteristics (Jun and Yi, 2020; Koay et al., 2022; Lee et al., 2021). The literature also explained that attitudes and purchase intentions are affected by the DI's attractiveness and by the resemblance between the DI and the brand (Cheah et al., 2024). Drawing upon the literature on social media and digital influencers (see Appendix A), the present study contributes to the existing literature in several significant ways. First, it extends the DIP model by examining how DIs' perceived influence, reputation, and trustworthiness affect CI in endorsed brands. Second, it advances the literature by investigating the impact of CI on customer attitudes toward endorsed brands, followed by the influence of subjective norms and PBC—key components of the TPB—in shaping customer intentions toward endorsed brands. Third, the study provides further insights by analyzing the relationship between customer purchase intentions and actual purchase behavior. Finally, this research offers a novel contribution by exploring the mediating role of CI in the relationship between DI's characteristics (trustworthiness, reputation, and perceived influence) and customer attitudes toward endorsed brands.

2.5. Hypotheses development

2.5.1. Trustworthiness and CI

Trust is stated as the degree of confidence in the speaker's message by the listeners and their degree of acceptance. Source trustworthiness is conceived once consumers believe that the communication source is honest, unbiased, and reliable (Baudier et al., 2023; Ohanian, 1990), along with the degree to which the audience believes the speaker's assertions are genuine (Wang et al., 2020). Trustworthiness is a key factor in social media influencer marketing (Masuda et al., 2022). Past studies have illuminated the significance of source trustworthiness in stimulating customer attitudes, behavioral intentions, and end behavior (Koay et al., 2022; Lou and Yuan, 2019). Furthermore, Li et al. (2015) asserted that trust significantly affects CI. The extant literature argues that customer trust in DIs and their promoted brands on digital platforms enhances CI, which consequently impacts customer attitude towards the advertised product websites (McMillan et al., 2003). DIs advocate brands to others based on their personal use experience (Uzunoğ;lu & Misci Kip, 2014), which improves consumer perceptions of trustworthiness (Schouten et al., 2020). The linkage between trustworthiness and CI is yet to be considered. This linkage can be justified with the help of Hovland et al. (1953), persuasion theory. According to this theory, the source characteristics relate to the quality of the persuasive information. Trustworthiness as a source characteristic can be argued to influence CI. Once an influencer establishes trust with their audience, then trustworthiness can act as a catalyst to amplify CI. Thus, based on the aforementioned arguments, it can be presumed that DIs' trustworthiness can positively influence CI in endorsed brands, thus it can be hypothesized that.

H1. DIs' trustworthiness positively impacts CI in endorsed brands.

2.5.2. Reputation and CI

DIs' reputation is a crucial component that impacts their followers' decisions. It is defined as the extent to which followers perceive the information shared by the influencer as reliable and genuine. Ryu and Han (2021) proposed communication skills, influence, authenticity, and knowledge as dimensions of reputation. DIs' creative and engaging content prompts their persuasive communication with followers, and their expertise with the endorsed brand helps them to be perceived as a reliable source of knowledge. Moreover, the content shared by influencers is carried out by millions of individuals, demonstrating the influencers' influence. Thus, the influencer recommendation about a product is perceived as genuine content (Ryu and Han, 2021). Information shared by reputable DIs increases individuals' involvement with the shared content, i.e., DIs endorsed brand, which sparks individual personal aspiration and enhances subjective value while seeking information (Balabanis and Chatzopoulou, 2019; Masuda et al., 2022). Further, according to Hovland et al. (1953), persuasion theory, reputation aligns with the contextual factors (persuasion situation) that can enhance or hinder message reception. According to the aforementioned arguments, in this study, the CI can be influenced by reputation. Thus, it can be assumed that DIs' reputation positively influences CI in endorsed brands. Therefore, we hypothesize.

H2. DIs' reputation positively impacts CI in endorsed brands.

2.5.3. Perceived influence and CI

Perceived influence explains how DIs affect followers' perspectives through their persuasion power (Jiménez-Castillo and Sánchez-Fernández, 2019). DIs' perceived influence is the extent to which individuals consider the information valuable and relevant to make the right purchase decisions (Crespo et al., 2015). By providing inspiring and engaging content, DIs raise their followers' interest, which ultimately leads to greater involvement in the message content (Kapitan and Silvera, 2016). Moreover, DIs' message information value, expertise, source credibility, and perceived similarity also increase perceived influence (Crespo et al., 2015). Additionally, consumers carefully evaluate the message and value recommendations obtained from DIs based on personal experience (Uzunoğ;lu & Misci Kip, 2014) and information value (Crespo et al., 2015). Further, according to Hovland et al. (1953), persuasion theory, a favorable persuasion situation can amplify message impact, like how an influencer's positive reputation boosts the effectiveness of the content. Consumer involvement corresponds to the persuasion object. The linkage between perceived influence and CI is yet to be considered; however, based on the aforementioned arguments, it can be assumed that DIs' perceived influence positively affects CI in endorsed brands. Therefore, we propose.

H3. DIs' perceived influence positively influences CI in endorsed brands.

2.5.4. CI in endorsed brand and attitude towards the endorsed brand

Involvement refers to a motivational state of mental perception, categorized as a goal-focused and goal-directed stimulus that arouses a consumption need for a certain item or service (Cheung and To, 2011). CI revolves around an individual's intrinsic needs, interests, and values (Zaichkowsky, 1985). Consumer research illustrates that involvement is an influencing element in the customer decision-making process (Engel et al., 1968). In relationship marketing, highly involved customers are more interested in associating with service providers (Varki and Wong, 2003). Customer involvement with brands has been suggested as a relevant factor in virtual advertising. The importance of a brand ad to its website setting is essential for high customer involvement (Humphrey Jr et al., 2017). Moreover, CI improves the association between consumer satisfaction levels, loyalty, and intention (Serravalle et al., 2023; Tata et al., 2019). Prior research also argued that customer involvement with the product and destination induces advertising usefulness and facilitates in development of consumer product-related mindsets and in generating their buying intention and eWOM (Abbasi et al., 2023; Wu et al., 2008). Thus, we assume that CI in an endorsed brand can lead to an attitude towards the endorsed brand. Hence, we postulated that.

H4. CI in endorsed brands positively influences the attitude towards the endorsed brands.

2.5.5. Customers' attitude towards endorsed brands, PBC, subjective norms, and purchase intention

According to the TPB, attitude, PBC, and subjective norms lead to behavioral intention (Ajzen, 1991; Song et al., 2012). Attitude refers to a once favorable or unfavorable assessment of performing a specific action

(Ajzen, 1991). Subjective norms replicate social norms or normative forces that may affect an individual's behavioral intention and include the perceived approval of social peers, for example, family and friends, on the particular behavior (Kim et al., 2016). The attitude represents the individual's evaluation of how favorable the intended behavior might be and is substantial in achieving an explicit behavior (Pillai et al., 2022). Alongside, PBC comprises the individual conviction to control their behavior, such as having sufficient abilities, skills, and resources to realize the behavior (Ajzen, 1991). Cho et al. (2015) advocated that the TPB is convenient for discovering the effect of psychological and social causes on consumer behavioral intention. Previous studies on digital media demonstrate a positive link between subjective norms, attitude, PBC, and consumer intention (Zahrai et al., 2022). Further, the linkage between attitude, subjective norms, PBC, and purchase intention is well-established in the literature (Kalam et al., 2025). This study aims to contribute to the discourse on DIs' role in digital marketing by examining these relationships in a specific context. Therefore, we propose the following hypothesis.

H5. Attitude towards the endorsed brands positively influences customer purchase intention.

H6. Subjective norms positively influence customer purchase intention.

H7. PBC positively influences customer purchase intention.

2.5.6. Purchase intention and actual behavior

Ajzen (1991) recommended that intention is a direct antecedent of actual behavior. Purchase intention is one of the most considered concepts in past literature. The association between customer intentions and the final purchasing behavior of a product has significantly attracted researchers' attention. Previous investigations have shown a positive link between purchasing intentions and behavior (Lin et al., 2022). In the social media context, the consumer's intention to co-create significantly impacts his/her co-creation behavior (Cheung and To, 2016). The current study assumes a positive association between customer purchase intention towards endorsed brands and actual behavior. Thus, we presumed.

H8. Purchase intention of customers positively influences their actual behavior.

2.5.7. Mediating role of CI

Zaichkowsky (1985) described CI as an individual's connection to an object based on their fundamental interests, intrinsic needs, and values. In the context of digital media, customer involvement with endorsed brands is mainly influenced by individual needs and values (Sarkar and Sarkar, 2019). DIs product experience and expertise in the digital platform through which they are endorsing the brand provide valuable information to customers in the form of recommendations and suggestions and facilitate them in making purchase decisions (Wang et al., 2020). This creates customers' positive brand perception by creating unique, informative, and engaging digital media content (Mcquarrie et al., 2013) and ensures CI, which in turn enhances customer desire to follow DIs' recommendations and develop a positive attitude toward digitally endorsed brands (Cheung et al., 2022). Further, perceived influential power exerted by DIs due to quality content and credibility generates customer engagement and raises their presumed value (Jiménez-Castillo and Sánchez-Fernández, 2019). Prior studies asserted that CI has a positive effect on attitudes toward either conventional mass media or net space advertising (Ko and Park, 2002), innovation performance (Cheng and Shiu, 2019), brand recall and recognition (Tsiotsou, 2013), information managing and search behavior (Andrews et al., 1990). Similarly, Belanche et al. (2020) examined the relationship between influencer product post fit and intention to interact with the influencer's account. Their findings indicate that CI has a significant moderating relationship with influencer product posts, particularly within the fashion product category.

Moreover, when consumers perceive DI as trustworthy, they become more receptive to the influencer's messages and develop a more favorable attitude toward the endorsed brand (Lou and Yuan, 2019). Their high involvement drives them to regularly watch, read, and comment on the influencer's posts, thereby reinforcing perception of the DI's trustworthiness and fostering a positive attitude (Ilieva et al., 2024). Furthermore, customers who are highly involved with DI are more inclined to critically assess influencer endorsements, particularly when the influencer has a strong reputation (Borges-Tiago et al., 2023). This CI strengthens their likelihood of forming a favorable attitude toward the endorsed brand. Additionally, according to the Elaboration Likelihood Model (ELM), highly involved consumers tend to engage in more systematic and thorough information processing (Masuda et al., 2022; Petty et al., 1986). If they perceive DI as powerful and influential, they are more likely to be drawn to the influencer's endorsements, ultimately leading to a favorable attitude toward the promoted brand.

Based on this discussion, it is evident that the mediating role of CI in the relationship between the DI characteristics—such as trustworthiness, reputation, and perceived influence—and customer attitudes toward endorsed brands remains underexplored in the existing literature and is yet to be unfolded. Therefore, we hypothesize that.

H9. CI in endorsed brands mediates the link between DIs' trustworthiness and attitude towards endorsed brands.

H10. CI in endorsed brands mediates the link between DIs' reputation and attitude towards endorsed brands.

H11. CI in endorsed brands mediates the link between DIs' perceived influence attribute and attitude towards endorsed brands.

3. Research methodology

3.1. Research setting and participants

This study used a cross-sectional research design to investigate the proposed relationships thoroughly. Utilizing the essentials of purposive sampling, which is regarded as the most credible sampling approach in marketing and social sciences research (Campbell et al., 2020; Etikan et al., 2016) and aids in assessing the pertinent respondents who may be trustworthy information providers (such as people who follow DIs on different social media sites and purchase the DIs-endorsed brands), the data were systematically collected from an emerging market, specifically Pakistan, to ensure a representative sample and inclusion of relevant demographics and features. The study focused on participants aged 17 to 40 actively engaging in the digital realm, managing profiles on prominent social media sites (e.g., Facebook, Instagram, YouTube, and TikTok. This specific age group was selected as they are more technologically proficient and actively use social media. Moreover, they frequently follow DIs, subscribe to their fan pages and channels, and engage with their content. Due to their frequent interactions with DIs, they are more likely to be influenced by their recommendations and inclined to purchase the brands they endorse. The participants were additionally categorized according to their subscription to various DIs' accounts, selected to represent a wide range of content areas and audience demographics. The investigation attempts to provide detailed insights into the dynamics of digital engagement and its potential repercussions within emerging markets by specifically focusing on this particular group of individuals.

A structured survey questionnaire was developed for data gathering, including respondents' demographic profiles and multiple-item scales adapted from previous studies. Before collecting data, we utilized the G*power tool and followed the guidelines of Faul et al. (2007) with specific input-parameters (e.g., $f^2 = 0.15$, α err prob = 0.05, power = 0.95, No. of predictors = 3) according to our study's model. This analysis identified that a minimum sample size of 119 is necessary for our study.

Data for this study were collected from various shopping malls in Islamabad, including Centaurus Mall and Giga Mall. The selection of multiple shopping malls in Islamabad was based on the city's status as the capital of Pakistan, which plays a crucial role in the national economy. Additionally, a significant proportion of the country's population resides in this region (Amir Zaib Abbasi et al., 2020a; 2020b). Participants were approached on weekends, as these days typically witness higher foot traffic, with many individuals visiting malls with their families for shopping. Specifically, individuals seated in the food court were targeted for participation. Potential respondents were approached and informed about the study, and only those who voluntarily agreed to participate were included. Prior to distributing the full questionnaire, screening questions were administered to ensure the suitability of participants. These questions included: (1) Do you have an account on social media platforms (e.g., Instagram, YouTube, Facebook, and TikTok)? (2) Do you follow DIs on these platforms? (3) Do you purchase products endorsed by DIs on social media? Only respondents who answered affirmatively to these questions were provided with the complete questionnaire. A total of 371 questionnaires were distributed, of which 322 valid questionnaires were used for final analysis, giving a response rate of 86.79 %. Out of the 371 questionnaires collected, 49 were discarded due to inauthentic responses, which included inconsistent patterns such as zigzag lines, straight-line responses, excessive neutral answers, and other irregular response patterns (Hussain et al., 2023, 2024). This step was undertaken to minimize bias and ensure the completeness and reliability of the data. To check for non-response bias, the final data were further divided into two categories such as early and late responses. Employing the Armstrong and Overton (1977) guidelines, an independent *t*-test was performed to compare early and late responses. The results indicated no significant differences between the two groups concerning the study's variables, suggesting that non-response bias was not a concern in this study. Table 1 presents a demographic summary of the participants.

Additionally, survey-based research that depends on self-reported data is at risk of common method bias (CMB) (Podsakoff et al., 2003). Our objective was to identify and mitigate potential CMB before verifying the reliability and validity of the data, as it was collected from a single source at a single point in time. For this purpose, the study has applied three approaches, such as Harman's one-factor exploratory factor analysis, and multicollinearity using the variance inflation factor – VIF (Kock, 2015). Furthermore, to mitigate sequencing effects and enhance the accuracy of the data, we implemented the randomized order of constructs in the questionnaire. VIF is the recent approach for calculating the CMB. In Harmen's single factor by Podsakoff et al. (2003), the threshold value of the total variance derived from a single factor needs to be less than 50 %. Whereas, all VIF values should be below the crucial threshold of 3.3 (Kock and Lynn, 2012) or 5 (Sarstedt et al., 2021), to confirm that there are no CMB issues in the study. The results from Harman's one-factor test indicated that all items explained 37.89 % of the model's variance, with all VIF values below 3.3. This suggests that the data is devoid of endogeneity and CMB issues.

Table 1
Demographics.

Gender	
• Male	57.5
• Female	42.6
Age	
• 17-24	56.2
• 25-40	43.8
Time Spent on the Internet	
• 1–2 h	8.4
• 3–4 h	22
• 5–6 h	37.6
• 7 or more hours	32
Time Spent on Social Media Platforms	
• 1–2 h	23
• 3–4 h	32.9
• 5–6 h	28
• 7 or more hours	16.1
Time Spent on Watching Videos	
• 1–2 h	56.2
• 3–4 h	21.1
• More than 4 h	22.7
Monthly Money Spent on Endorsed Brands	
• Less than 25000Pkr	81.4
• 25001-50000Pkr	9.9
• 50001-75000Pkr	4
• 75001Pkr or more	4.7

Table 2
Measurement model.

Construct-items	Loading	Cronbach's Alpha	rho_A	Composite Reliability	Average Variance Extracted (AVE)
DIT1	0.779	0.883	0.885	0.914	0.681
DIT2	0.827				
DIT3	0.831				
DIT4	0.853				
DIT5	0.833				
DIR1	0.867	0.804	0.817	0.884	0.718
DIR2	0.885				
DIR3	0.788				
DPI1	0.817	0.812	0.816	0.889	0.727
DPI2	0.867				
DPI3	0.874				
CI1	0.806	0.841	0.852	0.883	0.56
CI2	0.774				
CI3	0.793				
CI4	0.764				
CI5	0.706				
CI6	0.630				
ATB1	0.872	0.822	0.823	0.894	0.738
ATB2	0.866				
ATB3	0.839				
SN1	0.849	0.824	0.824	0.895	0.739
SN2	0.877				
SN3	0.853				
PBC1	0.881	0.835	0.839	0.901	0.752
PBC2	0.882				
PBC3	0.838				
PI1	0.868	0.847	0.849	0.907	0.765
PI2	0.891				
PI3	0.865				
APB1	0.735	0.841	0.847	0.883	0.556
APB2	0.732				
APB3	0.776				
APB4	0.758				
APB5	0.713				
APB6	0.760				

3.2. Measures

The questionnaire is comprised of 35 items related to 9 study constructs, adapted from prior papers. The questionnaire measured the perception of DIs' reputation (Hsu et al., 2013), trustworthiness (Ohanian, 1990), and perceived influence (Jiménez-Castillo and Sánchez-Fernández, 2019). Further, the questionnaire included measurement items on CI towards an endorsed brand from (Abbasi et al., 2023; Rodgers and Schneider, 1993) and TPB variables, i.e., the attitude towards the brand, subjective norms, and PBC measures from Ajzen (1991) and Yang et al. (2017), whereas purchasing intention from (Jiménez-Castillo and Sánchez-Fernández, 2019), and actual purchase behavior measures were modified from Lee (2009), see Appendix B, which includes the study's questionnaire. The items were assessed using a five-point Likert scale, ranging from 1 ("strongly disagree") to 5 ("strongly agree").

To evaluate the proposed relationships between the construct (Fig. 1), the Partial Least Squares (PLS) Structural Equation Modeling

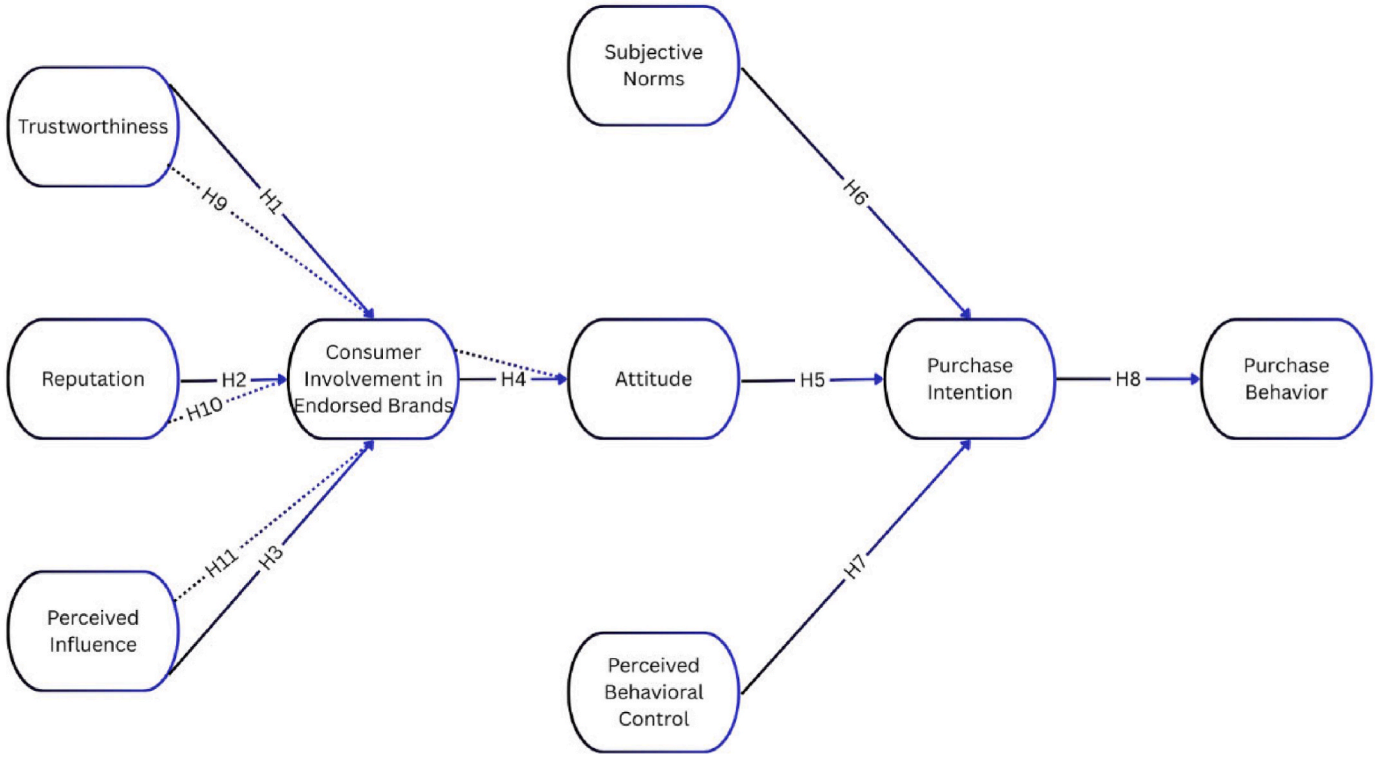


Fig. 1. Conceptual model integrating persuasion model and theory of planned behavior.

(SEM) approach was utilized. PLS-SEM is the most suitable approach for investigating the proposed model in this study, as the research aims to predict outcomes and extend or develop theory within a relatively novel context (Hair Jr et al., 2022). PLS-SEM offers solutions when the CB-SEM approach produces inadmissible outcomes or is unsuitable for complex models and small sample sizes. However, PLS-SEM provides greater flexibility in handling non-normal data and formative constructs (Hair et al., 2019). Thus, the current study has carried out the PLS-SEM analysis in SmartPLS 4.1.0.9 software. The evaluation followed a two-step approach: first, assessing the measurement model for reliability and validity tests, and second, performing bootstrapping with 10,000 resamples to determine the significance of proposed hypotheses (Hair Jr et al., 2022).

4. Data analysis

4.1. Measurement model evaluation

Evaluating the measurement model involves assessing the reliability and validity of the constructs. Reliability was determined through indicator reliability and internal consistency. According to standard criteria, all indicator loadings should exceed the recommended threshold of 0.70, indicating sufficient reliability (Hair et al., 2019). According to Hair Jr et al. (2022), the items with loadings between 0.5 and 0.7 do not need to be removed if they do not negatively impact convergent validity or internal consistency reliability. Additionally, Cronbach's alpha and composite reliability should exceed the threshold of 0.7 to confirm internal consistency. Further, Cronbach's alpha and composite reliability need to surpass the threshold values of 0.7, showing internal consistency (Hair Jr et al., 2022). The advised threshold value for average variance extracted (AVE) of all variables surpasses or equals 0.5 (Chin, 2009), representing the convergent validity. Table 2 illustrates that all the indicators of internal consistency, indicator reliability, and convergent validity surpassed the given threshold criteria.

Furthermore, to assess the discriminant validity, the Heterotrait and Monotrait (HTMT) method ratio was used. According to Henseler et al. (2015) discriminant validity is established when all constructs meet the threshold value of less than 0.85. In the present study, all HTMT values fall below this threshold, indicating no issues with discriminant validity,

Table 3
Discriminant validity.

	Apur	Att	CI	PBC	PInt	Pinf	Rep	Sub	Trust
Apur									
Att	0.608								
CI	0.652	0.823							
PBC	0.514	0.682	0.475						
Pint	0.609	0.677	0.7	0.552					
Pinf	0.52	0.781	0.737	0.604	0.618				
Rep	0.485	0.695	0.582	0.634	0.586	0.842			
Sub	0.557	0.785	0.595	0.779	0.529	0.698	0.589		
Trust	0.487	0.712	0.605	0.516	0.557	0.823	0.856	0.57	

HTMT values best under 0.85 and good at 0.90 or lower.
Note: *Apur-Actual purchase, Att-attitude, CI-Customer involvement in endorsed brands, PBC-perceived behavioral control, Pint-purchase intention, Pinf-perceived influence, Rep-reputation, Sub-Subjective norms.*

as shown in Table 3. Hence, reliability and validity are achieved. Further, the standardized root mean square residual (SRMR) was utilized to determine the model fit. The result showed that SRMR was less than the proposed maximum value of 0.08 (Benitez et al., 2020), suggesting model fit.

4.2. Structural model evaluation

The structural model was assessed to examine the proposed hypothesis by evaluating path coefficients, significance value, f^2, R^2, and Q^2 (Hair Jr et al., 2022; Henseler et al., 2016). Overall, the R-square values for CI in endorsed brand, attitude, purchase intention, and actual purchase were 40.4 %, 47.3 %, 35.2 %, and 27.4 %, respectively, suggesting that the proposed model had adequate explanatory power. Further, the values of Q^2 were found to be greater than 0, ranging from 0.146 to 0.346, while f^2 values for all accepted hypotheses were found above the suggested threshold of 0.02 (Cohen, 1992) (see Table 4), indicating that the predictive relevance of the model is acceptable (Hair Jr et al., 2022). Further, using the PLS-predict method, we examined out-of-sample prediction (Shmueli et al., 2016). The highest PLS (RMSE) values were lower than the LM (RMSE) values (Table 6), indicating a moderate level of predictive power. Overall, the results suggest that the research model possesses a strong predictive capability (Shmueli et al., 2019).

Direct Effects: Results using bootstrapping of 10000 samples demonstrate both direct and indirect correlations (Hair Jr et al., 2022). We evaluate the direct relationship of the path model through beta value, t-statistics, and level of significance. The results showed a significant positive influence of trustworthiness (H1–$\beta_{trust\rightarrow CI} = 0.171$, $t = 2.670$, $p < 0.01$), and perceived influence (H3–$\beta_{Pinf\rightarrow CI} = 0.472$, $t = 6.469$, $p < 0.001$) on CI in digital influencer endorsed brand. The results also revealed a significant positive impact of CI (H4–$\beta_{CIE\rightarrow Att} = 0.688$, $t = 17.437$, $p < 0.001$) on attitude. Further, attitude (H5–$\beta_{Att\rightarrow Pint} = 0.432$, $t = 5.806$, $p < 0.001$) PBC (H7–$\beta_{PBC\rightarrow Pint} = 0.196$, $t = 3.704$, $p < 0.001$) had a significant influence on purchase intention. The result also showed that purchase intention (H8–$\beta_{Pint\rightarrow PApur} = 0.524$, $t = 10.147$, $p < 0.001$) substantially influenced the actual purchase behavior. It is worth mentioning that reputation (H2–$\beta_{REP\rightarrow CI} = 0.041$, $t = 0.612$, $p > 0.05$) did not have a significant impact on CI, and subjective norms (H6–$\beta_{SUB\rightarrow Pint} = 0.036$, $t = 0.450$, $p > 0.05$) had an insignificant influence on purchase intention. Hence, the result showed that all hypotheses were supported, except H2 and H6. (Table 4 & Fig. 2).

Mediation Effect: In addition to testing direct relationships between the constructs, this research also examined the indirect effect of CI in the endorsed brand on the relationship between trustworthiness, reputation, perceived influence, and attitude towards the endorsed brand. The result suggested that CI significantly mediated the relationship between trustworthiness and attitude (H9–$\beta_{trust\rightarrow CI\rightarrow Att} = 0.118$, $t = 2.644$, $p < 0.005$), and also between perceived influence and attitude (H11–$\beta_{Pinf\rightarrow CI\rightarrow Att} = 0.325$, $t = 5.736$, $p < 0.001$). However, the mediating effect of CI on the relationship between reputation and attitude (H10–$\beta_{Rep\rightarrow CI\rightarrow Att} = 0.028$, $t = 0.606$, $p > 0.05$), was found to be insignificant (See Table 5).

5. Discussion

Regardless of the rising attention towards social media for promotional reasons in recent years, there is still a need to explore numerous facets of online media marketing research, such as the contemporary domain of DIs, which is still in its infancy (Jiménez-Castillo and Sánchez-Fernández, 2019). Particularly, prior studies explained the DIs as a source of eWOM on social media (Childers et al., 2019). The DI campaign is increasingly used in marketing promotion because it creates favorable reactions toward the endorsed brand. Still, there is not enough research in this direction to understand the process that makes DIs' endorsement effective (Torres et al., 2019). Therefore, the study aimed to present a DIP model by integrating DIs attributes, CI, and TPB in the context of brand endorsement. We examined different DIs attributes (perceived influence, trustworthiness, and reputation) on CI in endorsed brands, which further results in a positive attitude towards endorsed brands, their purchase intentions, and actual purchase behavior by extending the TPB. Most of our findings are correlated and support the proposed conceptual framework.

First, the findings showed that trustworthiness (H1) and perceived influence (H3) positively impact CI in endorsed brands. However, reputation showed an insignificant impact on CI (H2) in endorsed brands. This aligns with persuasion theory, which endorses the fact that message credibility and perceived expertise shape consumer attitudes and behaviors. Thereby, consumers are more likely to engage with influencer-endorsed brands when they perceive the endorser as credible and personally invested (Li et al., 2015). This is because consumers emphasize informational value and feedback from someone with personal experience who persuades them to make a purchase decision. Therefore, it is stated that both the perceived influence and trustworthiness of DIs play an important role in enhancing CI. These outcomes are stable and align with preceding studies (Jiménez-Castillo and Sánchez-Fernández, 2019; Masuda et al., 2022).

However, the insignificant effect of reputation (H2) on CI suggests that consumers may weigh authenticity and relatability over an influencer's overall reputation when interacting with endorsed products. This finding goes against source credibility models, which often assert reputation as a key driver of persuasion (Ohanian, 1990). A possible explanation for this finding can be attributed to the fact that today's digital consumers value trustworthiness and perceived expertise over reputation, in the context of influencer marketing. These results highlight the need to rethink how trust is built on social media, opening new opportunities for future research to understand trust-building mechanisms in social media.

CI (H4) was also found to significantly influence customer attitudes toward brands endorsed by DIs. This finding aligns with **persuasion theory**, which suggests that when consumers are highly involved, they are likely to analyze message content, brand attributes, and perceived authenticity of the DIs' endorsement. Consumer brand involvement enhances brand trust and develops a sense of connectedness and ownership, which improves customer attitude and perception of the endorsed products. The results are consistent with past studies outlining the importance of CI in the realm of brand choice and the satisfaction-loyalty paradox (Dagger and David, 2012; Humphrey Jr et al., 2017).

Table 4
Direct effects.

	Original Sample (O)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	P Values	f2	R2	Q2
Trust - > CI	0.171	0.064	2.670	0.004	0.02		
Rep - > CI	0.041	0.068	0.612	0.270	0.01		
Pinf - > CI	0.472	0.073	6.469	0.000	0.16	0.404	0.217
CI - > Att	0.688	0.039	17.437	0.000	0.90	0.473	0.346
Att - > PInt	0.432	0.074	5.806	0.000	0.16		
Sub - > PInt	0.036	0.081	0.450	0.326	0.01		
PBC - > PInt	0.196	0.064	3.074	0.001	0.04	0.352	0.263
PInt - > Apur	0.524	0.052	10.147	0.000	0.38	0.274	0.146

Table 5
Indirect effects.

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	P Values	Decision
Trust - > CI - > Att	0.118	0.122	0.045	2.644	0.004	Supported
Rep - > CI - > Att	0.028	0.03	0.047	0.606	0.272	Unsupported
Pinf - > CI - > Att	0.325	0.324	0.057	5.736	0.000	Supported

Table 6
PLSpredict.

Endogenous Construct	Items	PLS-RMSE	LM-RMSE	PLSRMSE - LMRMSE
Actual Purchase Behaviour	APB2	1.171	1.151	0.02
	APB4	1.1	1.124	−0.024
	APB3	1.045	1.053	−0.008
	APB1	1.015	0.941	0.074
	APB5	0.971	0.92	0.051
	APB6	1.074	1.085	−0.011

Furthermore, within the framework of the TPB, consumer involvement impacts perceived behavioral control and attitude, reinforcing the intention to develop positive brand associations.

Second, in the DIs brand endorsement context, results demonstrated partial support for the TPB model. The attitude and PBC (H5 and H7) were explained to have a substantial impact on buying intention. In line with the former work, our result suggested attitude and PBC as the most vital determinants of brand endorsement (Magrizos et al., 2021). From the TPB perspective (Ajzen,1991), positive attitudes towards an endorsed brand strengthen consumers' behavioral intentions, reinforcing their likelihood of purchasing the product. Given that DIs are considered **opinion leaders**, their endorsements serve as heuristic cues that shape consumer attitudes and promote trust. This suggests brands must strategically leverage DI personality traits and positioning to cultivate positive attitudes that translate into stronger purchase intentions.

Additionally, PBC was also observed to be a significant predictor of purchase intentions, as DI provides relevant product information and assurance about an endorsed product that boosts consumer confidence. This supports persuasion theory, which asserts the idea that consumers value reliable sources and informational influence when forming purchase decisions. These findings are consistent with previous studies (Chetioui et al., 2020; Chopra et al., 2021). Moreover, purchase intention (H8) also demonstrates a positive impact in predicting customer actual purchase behavior. The positive influence of purchase intention is grounded in the belief that the strong buying intention of customers, when shaped by DIs, will translate into real purchases. Customers act on their purchase intent because they believe that the influencer brand is more valuable and trustworthy. Moreover, customers are encouraged to follow their purchase intention when they receive social validity of the brand from an influencer's recommendation, leading to positive customer emotions and conversion of intentions into actual purchases. The findings are coherent with the prior studies investigating the role of purchase intention in a similar vein (Dang et al., 2020; Lin et al., 2022).

However, the subjective norm (H6) did not predict intentions. This finding aligns with the prior studies that have reported an insignificant influence of subjective norms on customer purchase intention in various contexts, including online fashion products (Widyarini and Gunawan, 2015), and green product purchasing (Ruangkanjanases et al., 2020). This result can be attributed to the increasing individual autonomy, which makes people more likely to adopt behaviors based on their attitudes and sense of control rather than on societal expectations. This tendency is particularly evident for products of a personalized or specialized nature, such as fashion items or environmentally friendly products, where consumers prioritize their intrinsic preferences and needs over societal norms. Furthermore, consumers today have greater access to various information sources, reducing their dependence on

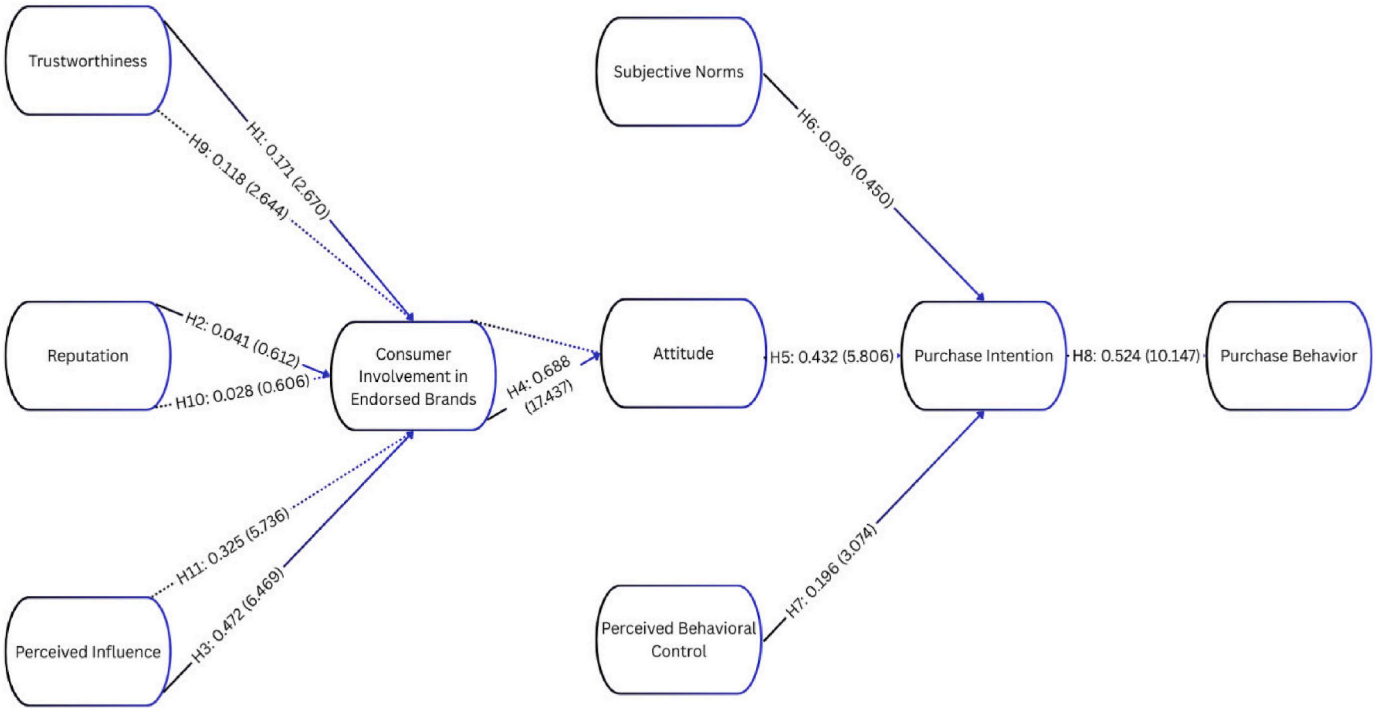


Fig. 2. Empirical validation model of the conceptual model.

social expectations. Additionally, the customer shift to online purchasing may also weaken the role of social norms, unlike traditional settings where peer influence is more immediate. This aligns with the Elaboration Likelihood Model (Petty et al., 1986) of persuasion, which suggests that when consumers process information via the central route of persuasion, they focus more on product attributes and self-relevance, rather than external social pressure. Therefore, DIs should focus more on the central route of persuasion by creating content that highlights personal experiences and authentic success stories rather than just societal expectations. This approach will bring in deeper engagement, as it connects with the customer on a personal level and brings in the desired change.

Third, the study results also exhibited that perceived influence and trustworthiness are key characteristics contributing to endorsement effectiveness, as they indirectly influence brand attitudes through the mediating effect of customer involvement (H9 and H11). Nonetheless, CI (H10) did not show a significant mediating effect concerning DIs' reputation and attitude. These results illustrate that trustworthiness and perceived influence can transmit their effect on attitudes via customer involvement. The mechanism of customer involvement can be strengthened by the transmittal effect of trustworthiness and perceived influence, which further leads to customer positive attitudes and purchase intention (Kim et al., 2018). These results are consistent with the notion of Hovland et al. (1953) persuasion theory, which emphasizes that the credibility factor acts as a driving force for attitudinal change.

Overall, our findings confirmed that DIs who are trustworthy and exert perceived influence could facilitate **customer involvement (CI).** Consumers seek product-related information from credible sources who have personally used the product and are perceived to be similar, as is the case with DI's. This further enhances customer brand-related attitudes, which in turn predict purchase intentions and actual purchase behavior. By combining the persuasion theory with TPB, this study provides deeper theoretical implications on brand endorsement by highlighting the psychological processes via which CI improves consumer-brand connections.

5.1. Theoretical implications

The study makes several important theoretical contributions. First, by integrating DIs' characteristics with persuasion theory and the TPB framework, this study brings an interdisciplinary perspective to marketing, psychology, and communication. This integration brings new perspectives to influencer marketing literature and provides a comprehensive understanding of how influencer attributes shape consumer behavior across multiple domains. Additionally, this study extends and strengthens the applicability of TPB and persuasion theory in a digital marketing and influencer-driven market landscape.

Second, while prior research has studied the effects of influencer characteristics such as perceived influence, emotional attachment, authenticity, trustworthiness, expertise, and attractiveness on consumer purchase intention, awareness, and loyalty toward the recommended brands (Jiménez-Castillo and Sánchez-Fernández, 2019; Jun and Yi, 2020; Koay et al., 2022; Lou and Yuan, 2019), rather than examining their combined impact within a theoretical framework. This research addresses this gap by examining the combined impact of influencer attributes within the framework of TPB and persuasion theory, introducing the DIP model. The DIP model explains how DIs psychologically and behaviorally influence consumer decisions. This offers a deeper understanding of how DIs' characteristics shape consumer involvement, ultimately affecting attitudes, subjective norms, perceived behavioral control, and purchase behavior. Given the growing impact of social media influencers on consumer decision-making, it is crucial to identify the specific attributes of DIs' content that drive consumer involvement, shape attitudes, and influence purchasing behavior. To address this, we introduce a comprehensive model that integrates key digital influencer attributes—perceived influence, trustworthiness, and reputation (Ki and Kim, 2019; Koay et al., 2022) with CI (Cheng and Shiu, 2019) and the TPB framework.

Our study empirically validates the interrelationships among these factors, demonstrating how trustworthiness, perceived influence, and reputation affect CI in endorsed brands, attitudes toward endorsed products, subjective norms, PBC, purchase intentions, and actual purchasing behavior. By synthesizing persuasion theory with the TPB, this research introduces the DIP model—an integrated framework that deepens our understanding of how influencer endorsements shape consumer behavior. This study makes a significant contribution by bridging influencer marketing, persuasion theory, and behavioral intention models, offering valuable insights for both scholars and practitioners seeking to optimize digital marketing strategies. Additionally, this study provides avenues for future research, encouraging scholars to examine how different influencer types (micro vs. macro) impact consumer attitudes and behaviors, the role of content authenticity, engagement style, and sponsorship transparency in influencing consumer trust and identification, and the moderating effects of industry type, product category, and consumer demographics on DI's effectiveness.

Third, the identification of consumer involvement in endorsed brands as a mediating mechanism between influencer characteristics, TPB variables, and purchase decisions is a key theoretical advancement. Previous studies have examined how influencer traits affect consumer behavior, but our research highlights how important consumer involvement is in explaining these relationships. The study suggests that just having an influencer who seems credible isn't enough; it comes down to how much consumers get involved, think about, and take in what the influencer is sharing. This insight increases our understanding of persuasion and how people make decisions, showing that when consumers are more involved, digital influencers have a greater impact on consumer choices.

Last, the study challenges prior assumptions by revealing that an influencer's reputation does not significantly affect consumer involvement, suggesting that trustworthiness and perceived influence are more critical determinants. This study raises a concern related to marketing models that rely mainly on reputation-based criteria such as number of followers, likes, shares, etc. Instead, the study argues to apply a consumer-focused approach that considers how cognitive and emotional involvement influences what consumers think and how they act. These insights suggest an agenda for future research on the evolving role of digital influencers in shaping consumer behavior and encourage researchers to explore how consumer involvement levels affect various digital marketing strategies and consumer outcomes.

5.2. Practical implications

The study also offers notable practical implications apart from theoretical relevance. Based on the findings of this study, we propose that brands should be very selective in choosing DIs, as influencer attributes affect consumer involvement, thereby affecting attitudes, subjective norms, perceived behavioral control, and purchase behavior. This is crucial as a mismatch between the product, consumer, and influencer trio can reduce the potency of influencer marketing (Duffek et al., 2025). Maintaining consumer–influencer-product congruence is a must for both brands and DIs. Choosing an influencer who is thought to have expertise in the endorsed brand and whose follower lifestyle matches the target audience can help in achieving this strategic alignment. This further ensures the endorsed brand resonates well with potential customers (Bastrygina et al., 2024). Further to evaluate an influencer's trustworthiness, brands can use specialized platforms that offer credibility scores based on data analytics and past performance metrics. For instance, platforms like HypeAuditor and CreatorIQ provide in-depth analyses of influencers' audience demographics, engagement rates, and authenticity indicators, aiding brands in making informed decisions (Eslami et al., 2024). By using these tools for the influencer

selection process, brands can increase the effectiveness of their marketing campaigns, ensuring better alignment with influencers who resonate authentically with their target audience.

The suggested DIP model demonstrates the significance of customer engagement in turning DI characteristics into favorable brand attitudes and purchase intent. This emphasizes that companies should move from passive endorsement to active consumer involvement to maximize the impact of influencer marketing. This is important as it ensures two-way communication between customers and influencers, strengthens emotional ties, and creates strong brand connections. Social media managers responsible for influencer alliances should thus understand how consumers interact with influencers and adjust their digital marketing strategies. This can be achieved by using analytics platforms like Upfluence to track engagement data such as likes, shares, and comments, and ascertain which content works best. Recognizing the types of influencer content that drive the highest consumer involvement is vital for maximizing influencer marketing effectiveness.

Moreover, it is also important to encourage DIs to create material that promotes consumer talk, increases community involvement, and feels more linked to them. This can be achieved with tailored responses to followers, honest disclosure regarding sponsorships, and user-generated content. Therefore, brands should analyze the influencer's content for quality, authenticity, and alignment with their brand's values. This approach ensures that partnerships resonate with target audiences and maintain brand integrity.

Finally, customer trust in DIs plays a critical role in increasing consumer involvement. Consequently, marketers should give the utmost importance to authenticity and openness in influencer relationships. This will help in building consumer loyalty and long-term partnerships. Incorporating these practical implications into their digital marketing strategies, a brand can leverage influencers for meaningful consumer-brand interactions for continuous engagement and buying behavior.

6. Limitations and future research directions

Despite the study's notable contribution to an existing body of literature, it has certain limitations that provide opportunities for future research. First, this research uses a cross-sectional design, limiting our ability to examine how consumer attitude changes over time. Future research should adopt a longitudinal approach to gain deeper insights into how consumer involvement and brand attitudes evolve over time.

Secondly, this study was conducted in Pakistan, a collectivist society. Therefore, future research should test the model in individualistic cultures (e.g., the U.S. or Western Europe) and explore Hofstede's dimensions to enhance generalizability. This will contribute to the scholarly understanding of variations in customer purchase behavior for the same product categories across diverse cultural contexts. Additionally, examining different types of endorsed brands (e.g., utilitarian and hedonic brands, or low and high-involvement brands) and influencer categories (e.g., nano, micro, macro, mega) could provide an understanding of variations in consumer responses. Moreover, future studies should examine the influence of DIs on customer purchase behavior across various social media platforms, such as X and Snapchat etc., to assess variations in the study findings based on different digital mediums.

Third, cultural and societal structures influence marketing effectiveness, emphasizing the need for cross-cultural studies essential for brands seeking to select suitable influencers. Consumer involvement may also vary across age groups; younger generations are more likely to be involved with digital influencers. Future research should use multi-group analysis to explore how DI characteristics impact purchase behavior across different cultures and age cohorts.

The study employed TPB to examine trustworthiness, perceived influence, and reputation on followers' behavioral outcomes. Future studies may further extend the DIP model to examine the direct impact of DIs' attributes (Ki et al., 2020; Koay et al., 2022) on the extended TPB variables, and the moderating effect of perceived risk and social media presence. Further, studies may explore influencers' attributes in other contexts, such as health and wellness, gaming and esports, and entertainment, with attention to their direct and indirect associations with customer well-being, aspirational behaviors, audience engagement, and the role of virtual influencers (Allal-Chérif et al., 2024; Dai et al., 2022; Oltra et al., 2022)

Furthermore, the study found an insignificant relationship between subjective norms and purchase intentions, suggesting a need for further research to explore social research mechanisms. Further studies may also examine other DIs characteristics such as to understand their impact on followers' decisions. Lastly, future research should also explore alternative theoretical frameworks, such as the Model of Goal-Directed Behavior (MGB) and the Elaboration Likelihood Model, to enhance the understanding of influencer marketing effectiveness.

CRediT authorship contribution statement

Muhammad Shahzeb Fayyaz: Writing – review & editing, Writing – original draft, Conceptualization. **Amir Zaib Abbasi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **Sanjeev Kumar:** Writing – review & editing, Writing – original draft, Methodology. **Ammar Qureshi:** Writing – original draft, Software, Investigation, Data curation. **Khalil Hussain:** Writing – original draft, Methodology, Investigation, Formal analysis. **Lakhi Muhammad:** Writing – review & editing, Writing – original draft.

Informed consent

Informed consent was obtained from all participants whom we reached out for data collection.

Ethics approval

All procedures performed in studies involving human participants were conducted following the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Declaration of competing interest

The authors confirm that there is no conflict of interest related to this study.

Appendix A. Digital media influencers' and content characteristics

Author(s)	Antecedents	Moderator (MOD)/Mediator (MED) Variables	Outcome variables	Context of the study
Magno and Cassia (2018)	• Information quality • Blogger trustworthiness	N/A	• Behavioral intention	• Social media influencers

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Author(s)	Antecedents	Moderator (MOD)/Mediator (MED) Variables	Outcome variables	Context of the study
Jiménez-Castillo and Sánchez-Fernández (2019)	• Perceived influence	• Brand engagement in self-concept (MED) • Brand expected value (MED)	• Intention to purchase recommended brand	• Digital influencers
Jin et al. (2019)	• Trustworthiness • Brand attitude • Envy	• -Social presence (MED) • Appearance-related self-discrepancy (MOD)	Celebrity condition (traditions vs Instagram)	• traditions vs Instagram information
Lou and Yuan (2019)	• Informative value • Entertainment value • Expertise • Trustworthiness • Attractiveness • Similarity	• Age, Gender and Involvement (MOD) • Trust in branded posts (MED) • Brand awareness (MED)	• Purchase intentions	• Influencer Marketing on social media
Ki et al. (2020)	• Inspiration • Visual aesthetics • Enjoyment • Similarity • Informativeness • Expertise	• Ideality needs fulfilment (MED) • Relatedness needs fulfilment (MED) • Competence needs fulfilment (MED) • Attachment (MED)	• Attachment transfer	• Social media influencers
(Jun and Yi, 2020)	• -Influencer interactivity • Influencer authenticity	• Brand trust (MED) • Emotional attachment (MED)	• Loyalty	• Social media influencers
Taillon et al. (2020)	• Similarity • Likeability • Attractiveness	• Closeness (MOD)	• Attitude Influencer • Purchase intentions • Word of mouth	• Social media influencers
Casaló et al. (2020)	• Originality • Uniqueness • Quality A • Quantity B	• Online interaction property (MOD) • Perceived with personal intentions (MOD) • Opinion leadership (MED)	• Intention to interact • Intention to recommend • Intention to follow the advice	• Influencers on Instagram
Sokolova and Perez (2020)	• Expertise • Credibility • Attractiveness • Social interaction • Entertainment • Information	• Parasocial relationship (MED) • Attitude towards YouTube video (MED) • Intention to watch fitness videos on YouTube (MED) • Attitude towards fitness (MED)	• Intention to exercise	• YouTube Influencers
Lee et al. (2021)	• Materialism	• Authenticity (MED) • Consumerism (MED) • Creative inspiration (MED) • Envy (MED) • Trust towards social media-related posts (MED)	• Purchase frequency	• social media influencers
Koay et al. (2022)	• Trustworthiness • Attractiveness • Expertise	• Materialism (MOD)	• Purchase intention	• Social media influencers
Zhang and Huang (2022)	• Influencer attraction • Government promotion	• Pandemic suspicion (MOD)	• Travel interests	• Influencer marketing
Abbasi et al. (2020a)	• Informativeness • Entertainment • Irritation • Credibility	• Advertising value (MED) • Attitude towards Vlog (MED)	• eWOM • POW • Customer referral • Destination visit intention	• Social media Vlogs
Han and Chen (2022)	• Source credibility	• User attitude (MED) • Influencer follower (MOD)	• Intention to visit	• social media
Pop et al. (2022)	• SNI Trust	• Desire (MED) • Information search (MED) • Evaluation alternative (MED) • Purchase decision (MED) • Satisfaction (MED)	• Experience sharing	• Social media influencers on travel
Masuda et al. (2022)	Personal attributes: • Attitude homophily, • Physical attractiveness, • Social attractiveness	Controlled variables: • Age • Gender • Product/Service • Influencer type Characterizations • Trustworthiness (MED) • Perceived expertise (MED) • Parasocial relationship (MED)	• Purchase intention	• Social media (YouTube) influencers
Leung et al. (2022)	• Self-disclosure • Self-prototypicality Prototype clarity	• Trust (MED) • Social attraction (MED) • Gender (MOD)	• Visit intention	• Social media influencers
Abbasi et al. (2022)	• Informativeness • Entertainment • Irritation • Credibility • Fantasy • Visual Appeal	• Adverting value (MED) • Place attachment (MED)	• Destination visit intention	• Social media Vlogs
Abbasi et al. (2023)	• Informativeness	• Social media image value (MED)	• eWOM	• Social media Vlogs

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Author(s)	Antecedents	Moderator (MOD)/Mediator (MED) Variables	Outcome variables	Context of the study
	• Entertainment • Irritation • Credibility • Personalization • Incentives	• Consumer engagement (MED) • Consumer Involvement (MED)		
Ooi et al. (2023)	• Mobile Convenience • Interactivity • Influence credibility	• Attitude towards the social media influencers (MED) • Attitude towards the product or service (MED) • Gender (MOD)	• Purchase behavior	• Social media
Hsieh et al. (2023)	• Influencer distinctiveness • Influencer similarity • Influencer prestige	• Influencer attractiveness (MED) • Follower influencer • Identification (MED) • Community identification (MED) • HBI phase (MOD)	• eWOM intention • Purchase intention	• Social media influencers
Shuqair et al. (2024)	• Facial expression (Broad vs slight smiles)	• Emotional contagion (MED) • Endorsement type (Utilitarian vs hedonic) (MOD)	• Consumer responses (engagement, purchase intention)	• Social media influencers

Note: *Some variables are positioned as mediators (MED), but indirect effects were not formally tested to establish mediation.*

Appendix B. The Study's Questionnaire

Digital Influencer Trustworthiness	
1	The digital influencer I follow for endorsed brands is dependable.
2	The digital influencer I follow for endorsed brands is honest.
3	The digital influencer I follow for endorsed brands is reliable.
4	The digital influencer I follow for endorsed brands is sincere.
5	The digital influencer I follow for endorsed brands is trustworthy.
Digital Influencer Reputation	
1	The digital influencer I follow for the endorsed brands has a reputation for being honest.
2	The digital influencer I follow for the endorsed brands has a good online reputation.
3	I believe many internet users recognize the digital influencer that I follow for endorsed brands.
Perceived Influence	
1	The digital influencer that I follow for endorsed brands offers security in my choice.
2	The digital influencer that I follow for endorsed brands helps me make better choices.
3	The digital influencer that I follow for endorsed brands gives me relevant information for my choice.
Customer Involvement in Endorsed Product Category	
1	The brands endorsed by digital influencers on social media are important to me.
2	I have a strong interest in brands endorsed by digital influencers on social media.
3	Brands endorsed by digital influencers on social media matter to me.
4	It would give me pleasure to purchase an influencer-endorsed brand for myself.
5	When I buy an influencer-endorsed brand, it feels like giving a gift to myself.
6	Owning an influencer-endorsed brand is a pleasure for me.
Attitude towards Brand	
1	Based on the digital influencer's information, I have a positive perception of the endorsed brand.
2	Based on the digital influencer's endorsement, I believe purchasing this brand would be enjoyable.
3	Based on the digital influencer's endorsement, I believe purchasing this brand would be a relaxing experience.
Subjective Norms	
1	My friends or relatives support my use of this brand endorsed by a digital influencer.
2	People who are important to me prefer that I use this brand endorsed by a digital influencer.
3	People whose opinions I value prefer that I use this brand endorsed by a digital influencer.
Perceived Behavioral Control	
1	If I want to purchase and use this endorsed brand, I can easily do so.
2	I have enough resources and knowledge to purchase and use this endorsed brand.
3	I find it easy to purchase and use this endorsed brand.
Purchase Intention	
1	I intend to purchase an endorsed brand based on the advice of digital influencers I follow.
2	I am willing to follow endorsed brand recommendations from the digital influencers I follow.
3	In the future, I plan to purchase the endorsed brands recommended by the digital influencers I follow.
Actual Purchase Behavior	
1	I often purchase this brand endorsed by a digital influencer.
2	I regularly purchase this brand endorsed by a digital influencer.
3	I purchase this brand because it is endorsed by a digital influencer.
4	I purchase this brand because I trust the digital influencer's endorsement.
5	I purchase this brand because it is recommended by a digital influencer I follow.
6	I purchase this brand because the digital influencer's endorsement assures its quality.

Data availability

Data will be made available on request.

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